

Anti-Derivatives

Derivatives and Anti-Derivatives (add $+C$ for general form)

$f'(x)$	$\Leftarrow$	$f(x)$	$\Rightarrow$	$F(x)$
0	$\Leftarrow$	1	$\Rightarrow$	x
1	$\Leftarrow$	x	$\Rightarrow$	$\frac{1}{2} x^2$
$2x$	$\Leftarrow$	x^2	$\Rightarrow$	$\frac{1}{3} x^3$
$n x^{n-1}$	$\Leftarrow$	x^n	$\Rightarrow$	$\frac{1}{n+1} x^{n+1}$
Multiply by power		New power is $(n+1)$		
New power is $(n-1)$		Divide by power		

$f'(x)$	$\Leftarrow$	$f(x)$	$\Rightarrow$	$F(x)$
$a \cos(ax)$	$\Leftarrow$	$\sin(ax)$	$\Rightarrow$	$-1/a \cos(ax)$
$-a \sin(ax)$	$\Leftarrow$	$\cos(ax)$	$\Rightarrow$	$1/a \sin(ax)$
$a e^{ax}$	$\Leftarrow$	e^{ax}	$\Rightarrow$	$1/a e^{ax}$
$2^x \ln(2)$	$\Leftarrow$	2^x	$\Rightarrow$	$2^x / \ln(2)$
$1/x$	$\Leftarrow$	$\ln(x)$	$\Rightarrow$	???

1. (Deriv. and Anti-Deriv.) Write derivative $f'(x)$ and anti-derivative $F(x)$ for the functions below.

A. $F(x) =$

$$f(x) = x^2 + x + 1$$

$$f'(x) =$$

D. $F(x) =$

$$f(x) = 4x^3 + 3x^2 + 2x + 1$$

$$f'(x) =$$

B. $F(x) =$

$$f(x) = 7x^4 + 5x^2 + 3$$

$$f'(x) =$$

E. $F(x) =$

$$f(x) = x^{1/2} + x^{2/3} + x^{3/4}$$

$$f'(x) =$$

C. $F(x) =$

$$f(x) = x^{5/2} + x^{1/2} + x^{-3/2}$$

$$f'(x) =$$

F. $F(x) =$

$$f(x) = e^{2x} + \sin(3x) + \cos(5x)$$

$$f'(x) =$$

2. (Initial Value Problems) Solve the initial value problems below.

A. Find $f(x)$ if $f'(x) = x + 1$ with $f(0) = 5$.

D. Find $f(x)$ if $f'(x) = 3x^2 + 1$ with $f(0) = 5$.

B. Find $f(x)$ if $f'(x) = x + 1$ with $f(1) = 5$.

E. Find $f(x)$ if $f'(x) = 3x^2 + 1$ with $f(1) = 5$.

C. Find $f(x)$ if $f'(x) = \cos(x)$ with $f(0) = 5$.

F. Find $f(x)$ if $f'(x) = \sin(x)$ with $f(0) = 5$.